Walks, paths, and connectivity from the adjacency matrix: a short account, with analogies

November 12, 2026

Abstract

For a finite simple undirected graph with adjacency matrix A, the entry $(A^k)_{ij}$ is the number of walks of length k from vertex n_i to vertex n_j (walks, not paths). A path is a walk with no repeated vertex. A graph on N vertices is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive. Each statement is proved and the test is run on two graphs.

1 Introduction

Let G be a finite simple undirected graph with N vertices $n_1, \ldots, n_N$ and adjacency matrix A (defined in section 2). The lecture board asks two questions about it, and the assignment adds a third [2, 3, 7].

(Q1) What does the entry $(A^k)_{ij}$ represent? The mark above A on the board is read as an exponent k [7]. *Answer:* it is the number of walks of length k from n_i to n_j (theorem 3.1); it counts walks, not paths.

(Q2) How can one tell from A whether G is connected? *Answer:* G is connected if and only if every entry off the diagonal of $B = A + A^2 + \cdots + A^{N-1}$ is positive (theorem 5.3).

(Q3) What is a path between n_i and n_j? *Answer:* a walk from n_i to n_j in which no vertex occurs twice (definition 4.1).

The answers come in the order Q1, Q3, Q2, because the test for Q2 needs the path. A full-text search of the course textbook [6] finds neither the walk-counting theorem nor a matrix test for connectivity, so both are taken from other sources, and every proof is written out in full; "derived" marks this paper's own proof of a result stated in a cited source. Textbook page numbers are printed ones, one lower than the PDF's.

Plain-words paragraphs precede the formal definitions, theorems, lemmas, and the worked example. They tell one story. A pond has *islands*, and some pairs of islands have a *bridge* between them (vertices and edges); two islands joined by a bridge are *next door*. A frog makes *trips* of *hops*, each hop crossing one bridge, and it may hop back or land twice on an island (a walk); a *fresh-island trip* lands on no island twice (a path), and the pond is *in one piece* when the frog can get from any island to any other (connected). A square *table* with a one in each cell whose two islands are next door, and a zero elsewhere, is the adjacency matrix.

2 Setting and definitions

Setting. $G = (V, E)$ is a graph in the sense of the textbook: a set $V = \{n_1, \ldots, n_N\}$ of vertices, $N \geq 1$, and a set E of edges [6, Definition 1.1.1, p. 9]. It is finite, simple (no edge from a vertex to itself, at most one edge between two vertices) and undirected (the edge joining

u and v, written uv, is the same as vu) [7]. The vertices u and v are *adjacent* if $uv \in E$ [6, Definition 1.1.4, p. 11]. The indices i, j, r run over $1, \ldots, N$; $k \geq 0$ is an integer.
In plain words. A pond has islands, and some pairs of islands have a bridge between them. Draw a square table with one row and one column for each island. Put a one in a cell when the two islands are next door, and a zero in every other cell.

Definition 2.1 (Adjacency matrix)**.** The *adjacency matrix* of G is the $N \times N$ matrix A with $A_{ij} = 1$ if n_i and n_j are adjacent and $A_{ij} = 0$ otherwise [6, p. 11]; the board states the same rule in other letters [2, 3].

Its entries are 0 or 1. Matrix powers are the usual ones: $(XY)_{ij} = \sum_{r=1}^{N} X_{ir} Y_{rj}$, $A^0 = I$ (the identity matrix) and $A^{k+1} = A^k A$. Every entry of A^k is a non-negative integer, since sums and products of non-negative integers are non-negative integers (*used in* theorem 5.3).
In plain words. A frog sits on an island and hops over a bridge to a next-door island. It may hop back, or land on the same island again. The islands it visits, in order, form a trip. A frog that never hops makes a trip with no hops.

Definition 2.2 (Walk)**.** A *walk* from u_0 to u_k is a sequence of vertices $u_0 u_1 \cdots u_k$ in which each vertex is adjacent to the one before it; its *length* is k, the number of edges used, each counted as often as it is used. Vertices and edges may repeat. This is the textbook's Definition 1.2.10 [6, p. 21], which lists edges as well; in a simple graph the vertices fix the edges. This paper also allows $k = 0$: a single vertex is a walk of length 0 from itself to itself. This is the paper's own convention; the textbook's usage assumes it [6, pp. 23, 49].

In plain words. A fresh-island trip lands on no island twice. So it never crosses the same bridge twice either. Every fresh-island trip is a trip, but a trip may land on an island twice.

Definition 2.3 (Path)**.** A *path* is a walk whose vertices are pairwise distinct. This is the course convention [7]. It is the textbook's *simple path* [6, Definition 1.2.13, p. 22]; the textbook's own word "path" means a walk with no repeated edge, and this paper never uses the word in that sense.

In plain words. A pond is in one piece when the frog can get from any island to any other. The way may cross many bridges. It is enough to ask for a fresh-island trip between each two islands.

Definition 2.4 (Connected)**.** G is *connected* if for any two vertices u, v there is a path from u to v. The board states this [2, 3]; the textbook's Definition 2.1.1 [6, p. 48] is the same with the textbook's own "path". A vertex is joined to itself by the path of length 0, so only pairs of different vertices matter.

Board graph. The board's matrix has rows 0 1 0, 1 0 1, 0 1 0 [2, 3]:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

It encodes the graph with edges $n_1 n_2$ and $n_2 n_3$, which is the example graph throughout. The board's drawing shows a third line that the matrix does not encode; the matrix defines the example.

3 Answer to Q1: the entries of A^k count walks

In plain words. Take the island table, and use as many copies of it as the trip has hops. Each cell then counts the trips with exactly that many hops from its row island to its column island. These trips may land on an island twice, so they are not all fresh-island trips.

Theorem 3.1 (Walk counting)**.** *For every integer $k \geq 0$ and all i, j, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j.*

The University of Galway notes state this for $k \geq 1$ with an induction proof [4, Theorem 1.1.3, p. 4]; the case $k = 0$ is this paper's convention. The proof is derived here.

Proof. By induction on k.

Base case $k = 0$. $A^0 = I$. A walk of length 0 from n_i to n_j is a single vertex that is both start and end, so there is exactly one such walk if $i = j$ and none if $i \neq j$. This is I_{ij}.

Inductive case. Assume that for some $k \geq 0$ and all i, r the entry $(A^k)_{ir}$ is the number of walks of length k from n_i to n_r. Fix i and j. Since $A^{k+1} = A^k A$,

$$(A^{k+1})_{ij} = \sum_{r=1}^{N} (A^k)_{ir} A_{rj}. \tag{1}$$

A walk $u_0 \cdots u_k u_{k+1}$ of length $k+1$ from n_i to n_j has $u_k = n_r$ for exactly one r; sort the walks by this r. Fix r. If n_r is adjacent to n_j (so $A_{rj} = 1$), the walks of this kind correspond one to one to the walks of length k from n_i to n_r: deleting the final vertex n_j gives such a walk, and appending n_j to such a walk gives a walk of the kind, because n_r is adjacent to n_j. By the hypothesis there are $(A^k)_{ir} = (A^k)_{ir} A_{rj}$ of them. If n_r is not adjacent to n_j (so $A_{rj} = 0$), no walk can end with the step from n_r to n_j, so there are $0 = (A^k)_{ir} A_{rj}$ of them. Every walk is counted for exactly one r, so the number of walks of length $k+1$ from n_i to n_j is $\sum_r (A^k)_{ir} A_{rj}$, which is $(A^{k+1})_{ij}$ by eq. (1). □

Warning: walks, not paths. The proof uses only that consecutive vertices are adjacent, so repeated vertices are counted. For a path, appending n_j may repeat a vertex and the one to one correspondence fails. In the board graph $(A^2)_{11} = 1$ and the walk counted is $n_1 n_2 n_1$, which repeats n_1; the only path from n_1 to n_1 is the single vertex n_1, of length 0 (a path of length $k \geq 1$ from a vertex to itself would repeat it). So "$(A^2)_{11}$ counts the paths of length 2 from n_1 to n_1" is false, although "counts the walks" is true. Also $(A^4)_{22} = 4$ by script, so the entry is not "1 if n_j can be reached from n_i".

4 Answer to Q3: the path between n_i and n_j

In plain words. Choose two islands. A fresh-island trip between them starts on the first island and ends on the second. A frog that stays put makes such a trip from an island to itself.

Definition 4.1 (Path between n_i and n_j)**.** A *path between n_i and n_j* (also a path *from n_i to n_j*) is a sequence $u_0, u_1, \ldots, u_k$ with $k \geq 0$, $u_0 = n_i$, $u_k = n_j$, in which u_{t-1} and u_t are adjacent for $1 \leq t \leq k$ and the vertices $u_0, \ldots, u_k$ are pairwise distinct. Its length is k.

This is definition 2.3 with the ends fixed, the textbook's Definition 1.2.13 with ends n_i and n_j [6, p. 22]. If $i \neq j$ such a path has at least one edge; if $i = j$ the only path is (n_i), since $u_0 = u_k$ with $k \geq 1$ would repeat a vertex. In the board graph $n_1 n_2 n_3$ is the only path from n_1 to n_3.

5 Answer to Q2: a connectivity test from A

The test rests on two facts about walks and paths. Both are stated in Grinberg [1, Corollary 2.9.7, p. 39, Section 2.9.4, p. 40] and the first is the textbook's Lemma 1.2.1 for $u \neq v$ without a length bound [6, p. 23]; the proofs are derived here.

In plain words. Suppose a frog gets from one island to another and lands on some island twice. Cut out all the hops between the two visits to that island. What is left is still a trip

between the same two islands, with fewer hops. Cutting again and again ends in a fresh-island trip.

Lemma 5.1 (Every walk contains a path)**.** *Let $W = u_0 \cdots u_k$ be a walk from u to v. Then there is a path from u to v of length at most k.*

Proof. By strong induction on k: for walks of length k we assume the statement for all shorter walks. *Base case* $k = 0$. $W = (u)$ has one vertex, so it is a path of length 0. *Inductive case* $k \geq 1$. If the vertices of W are pairwise distinct, W is a path of length k. Otherwise there are positions $s < t$ with $u_s = u_t$. Let $W' = u_0 \cdots u_s u_{t+1} \cdots u_k$ (if $t = k$, $W' = u_0 \cdots u_s$). It is a walk, because the only new consecutive pair is u_s, u_{t+1} and $u_s = u_t$ is adjacent to u_{t+1}. It runs from u to v and has length $k - (t - s) < k$. By the hypothesis W', and so W, yields a path from u to v of length at most $k - (t - s) \leq k$. □

In plain words. Pick two different islands and look at their cell in the tables for one hop, two hops, and so on. Go up to one fewer hops than there are islands. The frog can get from one island to the other exactly when one of these cells is not zero.

Lemma 5.2 (Reachability from the powers)**.** *Let $i \neq j$. There is a path from n_i to n_j if and only if $(A^k)_{ij} > 0$ for some k with $1 \leq k \leq N - 1$.*

Proof. ($\Rightarrow$) Let a path from n_i to n_j have length k. As $i \neq j$, $k \geq 1$. A path of length k has $k + 1$ pairwise distinct vertices among the N vertices of the graph, so $k \leq N - 1$. A path is a walk, so by theorem 3.1 $(A^k)_{ij} \geq 1$. ($\Leftarrow$) If $(A^k)_{ij} > 0$ with $1 \leq k \leq N - 1$, then by theorem 3.1 a walk of length k joins n_i to n_j, and by lemma 5.1 a path joins them. □

In plain words. Add the tables for one hop, two hops, and so on, up to one fewer hops than there are islands. Look only at cells for two different islands. The pond is in one piece exactly when none of these cells is zero. How large a cell is does not matter.

Theorem 5.3 (Sum test)**.** *Let $N \geq 2$ (for $N = 1$ the graph is connected) and $B = A + A^2 + \cdots + A^{N-1}$. Then G is connected if and only if $B_{ij} > 0$ for all $i \neq j$.*

This is Theorem 2 of Paramadevan and Sotheeswaran [5, p. 17]; the proof here is derived.

Proof. Each A^k has non-negative entries, so $B_{ij} = \sum_{k=1}^{N-1} (A^k)_{ij}$ is a sum of non-negative terms, and it is positive exactly when at least one term is positive. For $i \neq j$ this holds exactly when a path joins n_i and n_j, by lemma 5.2. So $B_{ij} > 0$ for all $i \neq j$ exactly when every pair of different vertices is joined by a path, which is definition 2.4. □

The cutoff $N - 1$ is the largest length a path can have (proof of lemma 5.2). The entries of B are walk counts, not path counts; only whether they are zero is used.

6 Worked example and limits

In plain words. We check the answer on two ponds with three islands. In the first, the islands lie in a row and the frog can get anywhere. In the second, two islands share a bridge and the third island stands alone.

All matrices and walk lists below were computed by the scripts in the `scripts/` folder, which also holds their outputs.

Connected: the board graph ($N = 3$, edges n_1n_2, n_2n_3). Here $B = A + A^2$ with

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

By theorem 3.1, $(A^2)_{11} = 1$ counts the walk $n_1n_2n_1$, $(A^2)_{13} = 1$ counts $n_1n_2n_3$, $(A^2)_{22} = 2$ counts $n_2n_1n_2$ and $n_2n_3n_2$, and $(A^2)_{12} = 0$ because no walk of length 2 joins n_1 to n_2. The script also gives $(A^3)_{12} = 2$, for the walks $n_1n_2n_1n_2$ and $n_1n_2n_3n_2$, and $(A^4)_{22} = 4$. All off-diagonal entries of B are positive, so G is connected by theorem 5.3: the pairs n_1n_2 and n_2n_3 get a positive entry from A, and the pair n_1n_3 gets $B_{13} = 1$ from A^2, which counts the path $n_1n_2n_3$.

Disconnected: the edge n_1n_2 and the isolated vertex n_3 ($N = 3$). Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The entries $B_{13} = B_{23} = B_{31} = B_{32} = 0$, so G is not connected by theorem 5.3. This agrees with the graph: no walk joins n_3 to n_1 or n_2.

Limits. The test answers the question using A alone, but it has limits. It reads only whether entries are zero: a positive entry says a walk of exactly length k exists, hence a path of length at most k, not a path of length exactly k. The diagonal of B is ignored, and a connected graph can have zeros there: for $N = 2$ and one edge, $B = A$ has a zero diagonal. The sum must run to A^{N-1}: on the path graph with edges n_1n_2, n_2n_3, n_3n_4 ($N = 4$) the script gives $(A^1)_{14} = (A^2)_{14} = 0$ and $(A^3)_{14} = 1$, so a sum stopped at A^2 would call this connected graph disconnected. The test is proved only for finite simple undirected graphs.

References

[1] Darij Grinberg. *An introduction to graph theory (Text for Math 530, Drexel University)*. arXiv:2308.04512v3 [math.HO], version of June 7, 2025. 2025. URL: `https://arxiv.org/abs/2308.04512` (visited on Nov. 6, 2026).

[2] *Lecture whiteboard photo 1*. Local file `sources/whiteboard-1.jpg`.

[3] *Lecture whiteboard photo 2*. Local file `sources/whiteboard-2.jpg`.

[4] *Linear algebra notes, 1.1 The Adjacency Matrix (University of Galway notes)*. University of Galway, lecture notes. The file carries no author name and no date. URL: `https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf` (visited on Nov. 6, 2026).

[5] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *JSc EUSL* 12.1 (2021). Eastern University, Sri Lanka, pp. 13–21. URL: `https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf` (visited on Nov. 6, 2026).

[6] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Graph Theory, study guide*. Local file `Grafneri_tesutyun.pdf`, in Armenian. 2015.

[7] *Summary of user statements*. Local file `sources/user-statements.md`. 2026.